9.8 NUCLEAR FUSION

When two or more light nuclei (A<20) combine to form a heavier nucleus, then it is called nuclear fusion. In the nuclear fusion, the mass of the resultant nucleus is less than the sum of the masses of original light nuclei. The mass difference appears as energy. The nuclear fusion never occurs at room temperature unlike nuclear fission. It is because when two light nuclei come closer to combine, they is strongly repelled by the coulomb repulsive force.

To overcome this repulsion, the two light nuclei must have enough kinetic energy to move closer to each other such that the nuclear force becomes effective. This can be achieved if the temperature is very much greater than 10^7 K. When the surrounding temperature reaches around 10^7K, lighter nuclei start fusing to form heavier nuclei and this resulting reaction is called thermonuclear fusion reaction.

Energy generation in stars:

The natural place where nuclear fusion occurs is the core of the stars, since their temperature is of the order of 10^7K. In fact, the energy generation in every star is only through thermonuclear fusion. In most of the stars including our Sun hydrogen atoms fuse into helium and in some stars helium atoms fuse into heavier elements.

The early stage of a star is in the form of cloud and dust. Due to their own gravitational pull, these clouds fall inward. As a result, its gravitational potential energy is converted to kinetic energy and finally into heat. When the temperature is high enough to initiate the thermonuclear fusion, they start to release enormous energy which tends to stabilize the star and prevents it from further collapse.

The sun's interior temperature is around $1.5\times10^7\ K$. In sun, $6\times10^{11}\ kg$ of hydrogen is converted into helium every second and sun has enough hydrogen such that these fusion reactions last for another 5 billion years. When the hydrogen is burnt out, the sun will enter into new phase called red giant where helium will fuse to become carbon. During this stage, sun will expand greatly in size and all its planets will be engulfed in it.

According to Hans Bethe, the sun is powered by **proton-proton cycle** of fusion reaction. This cycle consists of three steps and the first two steps are as follows:

$$ {}_1^1H + {}_1^1H \rightarrow {}_1^2H + e^+ + \nu \qquad (9.44) $$

$$ {}_1^1H + {}_1^2H \rightarrow {}_2^3He + \gamma \qquad (9.45) $$

A number of reactions are possible in the third step. But the most dominant one is

$$ {}_2^3He + {}_2^3He \rightarrow {}_2^4He + {}_1^1H + {}_1^1H \qquad (9.46) $$

The overall energy produced in the above reactions is about 27 MeV. The radiation energy we receive from the sun is due to these fusion reactions.

Elementary particles:

An atom has a nucleus surrounded by electrons and the nucleus is made up of protons and neutrons. Till 1960s, it was thought that protons, neutrons and electrons are fundamental building blocks of matter. In 1964, physicists Murray Gellman and

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